

Short Report

Networking Devices and Initial Configuration

Prepared by Raj Tiwari

Course completion note: The uploaded certificate confirms successful completion of "Networking Devices and Initial Configuration" through Cisco Networking Academy on 12 Apr 2026.

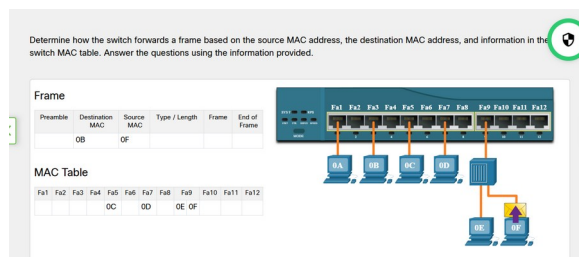
Overview. This course introduced the basic working of networking devices and the first steps required to configure and verify them. The activities helped me connect theory with hands-on practice by showing how devices such as switches and end hosts communicate inside a LAN.

1. Key concepts learned

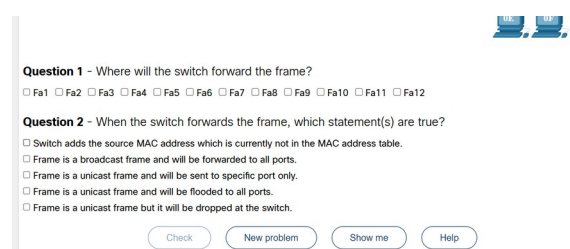
- **Role of networking devices:** I understood the purpose of switches, end devices, and ports in connecting systems inside a network.
- **MAC addressing:** I learned that each device has a unique MAC address used by switches to make forwarding decisions at Layer 2.
- **MAC address table:** The switch stores learned source MAC addresses against specific ports, which helps it send frames more efficiently.
- **Frame forwarding logic:** I practiced how a switch handles known unicast, unknown unicast, broadcast, and the importance of the incoming port.
- **Initial device configuration:** The course strengthened the basics of setting up network devices, checking interfaces, and understanding initial connectivity.
- **Verification mindset:** I learned to read the given topology, source and destination MAC addresses, and then decide how the frame travels in the network.

2. Screenshots of lab activities

The following screenshots are from my lab-based practice. They demonstrate switch forwarding behavior using source MAC, destination MAC, and the MAC address table.



Lab Screenshot 1: Frame details, MAC table, and network topology used to decide switch forwarding.



Lab Screenshot 2: Answer-selection interface for identifying the forwarding ports and frame behavior.

3. Practical understanding

- I understood that a switch first checks the source MAC address and learns it on the port from which the frame arrives.

- I learned that if the destination MAC address is not found in the MAC table, the frame is treated as an unknown unicast and is forwarded only through relevant other active ports, not back to the incoming port.
- The lab improved my ability to analyze port-by-port forwarding decisions instead of guessing, which is important for real troubleshooting.
- This practical work gave me a clearer understanding of how communication happens inside a local area network and how switches reduce unnecessary traffic once MAC entries are learned.

Conclusion. Overall, the course gave me a strong foundation in networking device behavior, especially switch operation and initial configuration. The lab exercises made the concepts easier to understand because I could apply MAC learning and frame forwarding rules in a real example.